Assume the contrary, i.e., 1 is an eigenvalue of c and let v be a corresponding eigenvector:

$$\begin{aligned} cv = v &\implies s_1 \dots s_n v = v \\ &\iff s_2 \dots s_n v = s_1 v. \end{aligned}$$

But since s_i only changes the ith coordinate of v, we get

$$s_1 v = v \quad \text{and} \quad s_2 \dots s_n v = v.$$

Repeating the same procedure, we get

$$s_i v = v$$

for all i. But this means

$$B(v, \alpha_i) = 0$$

for all i, and since B is nondegenerate, we get $v = 0$. But this is a contradiction, since v is an eigenvector. □

6.8. Proof of Gabriel's theorem

Let V be an indecomposable representation of Q. We introduce a fixed labeling $1, \dots, n$ on Q, such that $i < j$ if one can reach j from i. This is possible, since we can assign the highest label to any sink, remove this sink from the quiver, assign the next highest label to a sink of the remaining quiver, and so on. This way we create a labeling of the desired kind.

We now consider the sequence

$$V^{(0)} = V,\ V^{(1)} = F_n^+ V,\ V^{(2)} = F_{n-1}^+ F_n^+ V, \dots.$$

This sequence is well defined because of the selected labeling: n has to be a sink of Q, $n-1$ has to be a sink of $\overline{Q_n}$ (where $\overline{Q_r}$ is obtained from Q by reversing all the arrows at the vertex r) and so on. Furthermore, we note that $V^{(n)}$ is a representation of Q again, since every arrow has been reversed twice (since we applied a reflection functor to every vertex). This implies that we can define

$$V^{(n+1)} = F_n^+ V^{(n)}, \dots$$

and continue the sequence to infinity.